

Weiran Wang

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Research Interests

My research interests straddle the complex domain of networked systems and their interdisciplinary interfaces. On the one hand, I am interested in the network performance, protocols, and virtualization that underpin the essence of internetworking. On the other hand, I am willing to delve into the interconnected areas of distributed systems, machine learning, queuing theory, and game theory, highlighting the convergence of these disciplines with networked systems.

Education

Royal Institute of Technology (KTH)

Stockholm, Sweden

M.Sc in ICT Innovation - track Cloud and Network Infrastructures

Aug. 2020 - Mar. 2023

- **GPA:** 4.56/5, **Main Courses:** Advanced Internetworking, Queuing Theory, Advanced Distributed Systems, etc.
- **Master Thesis:** "Analysis of Flow Prolongation Using GNN in FIFO Multiplexing System"
- **Advisors:** Prof. Jean-Yves Le Boudec (EPFL), Prof. Viktoria Fodor (KTH) and Dr. Hossein Tabatabaee (EPFL)

Swiss Federal Institute of Technology in Lausanne (EPFL)

Lausanne, Switzerland

Master Study Exchange in the School of Computer and Communication Sciences

Sep. 2021 - Aug. 2022

- **GPA:** 5.26/6, **Main Courses:** Machine Learning, TCP/IP Networking, Network Calculus, etc.
- Semester and Degree Project done in Laboratory for Communications and Applications 2
- **Advisors:** Prof. Jean-Yves Le Boudec and Dr. Hossein Tabatabaee

Dalian University of Technology

Dalian, China

B.Eng in Software Engineering

Sep. 2016 - Jun. 2020

- **GPA:** 85.7/100, **Main Courses:** Mathematics, Computer Network, Computer Organization and Structure, Database System, etc.
- **Bachelor Thesis:** "The Research on Virtual Function Scheduling of Edge Network Based on Game Theory"
- **Advisors:** Prof. Zichuan Xu and Dr. Qiufen Xia
- **Extracurricular:** Attended the National Model United Nations Conference in the headquarters of UN in New York, the USA

Experience

Swiss Federal Institute of Technology in Zurich (ETHz)

Zurich, Switzerland

Research Assistant in Sustainable Networking

Oct. 2024 - Jun. 2025

- **Supervisors:** Prof. Laurent Vanbever, Dr. Romain Jacob

Ericsson

Stockholm, Sweden

5G Baseband Software Developer

Sep. 2022 - Oct. 2024

- Engaged with virtualized Distributed Units on UE/Cell Setup and Uplink Carrier Aggregation, which merges traditional 5G RAN with new Cloud RAN features.
- Served on the Module Product Care for User Plane Control Uplink, aiming to enhance the robustness of 5G baseband source code.
- Developed a framework for Software Quality Indications to track quality improvement initiatives systematically.
- Improved code quality and identified potential bugs by using static and dynamic code analysis tools.
- Participated in Ericsson's internal Hackathon with the topic of "Save 1 Watt in Baseband".

Ericsson

Stockholm, Sweden

Summer Internship R&D

Jun. 2021 - Aug. 2021

- Developed a tool to automatically check Ericsson code disclaimers of the created and last-modified dates from the git log.
- Extracted real-time Baseband Uplink Products data with test use cases, and visualized data on the Ericsson developer website.
- Experimented with the FOSS 'Include What You Use' with the Ericsson compiler to enable support for Ericsson's proprietary code.

Publication

Fantastic Joules and Where to Find Them. Modeling and Optimizing Router Energy Demand

Romain Jacob, Lukas Rölli, Jackie Lim, Jonathan Chung, Maurice Béhanzin, **Wang, Weiran**, Andreas Hunziker, Theodor Moroianu, Seyedali Tabaeiaghdaei, Adrian Perrig

ACM Internet Measurement Conference (IMC 2025), 2025

Research Projects

Incentivize Smarter Network Usage and Reduce Peak Load via Pricing Models

Zurich, Switzerland

ETH Zurich

Dec. 2024 - June 2025

- Evaluated how some alternative pricing schemes can incentivize users to reschedule their workloads, encouraging more balanced and sustainable network usage patterns.
- The ultimate goal is to identify pricing policies that effectively flatten system-wide throughput, thereby mitigating the need for continuous network infrastructure expansion.

Routers Datasheet Analysis

Zurich, Switzerland

ETH Zurich

Oct. 2024 - Nov. 2024

- Investigated the questions ‘Are datasheets representative of router power consumption in practice?’ and ‘Do these representations confirm the power efficiency improvements claimed by vendors?’
- Automated the extraction of power and bandwidth data using the GPT-4o API from official webpages, covering 777 routers from Cisco, Arista and Juniper.

Analysis of Flow Prolongation Using GNN in FIFO Multiplexing System

Lausanne, Switzerland

EPFL

Mar. 2022 - Dec. 2022

- Reproduced the GNN model based on PMOO and achieved an accuracy of 65% compared to 69.6% in the reference paper.
- Integrated the NetCal/DNC into topologies so that delay bounds can be calculated automatically once new networks are generated.
- Generated a novel dataset comprising over 160,000 topologies, each with the average of 25 servers and 115 flows, which was used for the adversarial attack purpose to simulate the network disturbance and jitter.
- Implemented FGSM attack to the network topologies with flow prolongations predicted by GNN.

Flow Analysis on GNN-Oriented Flow Prolongation

Lausanne, Switzerland

EPFL

Oct. 2021 - Feb. 2022

- Investigated the usage of NetCal/DNC with network topologies settings to compute the delay bound of the flow of interest.
- Investigated the usage of GNN to predict flow prolongations in new network topologies (trained GNN based on LUDB and PMOO).
- Analyzed the tightness of delay bounds and benchmarked the execution time computed by LUDB-FF, PMOO, TFA and SFA with PLP in source-sink tandem networks.

The Research on Virtual Function Scheduling of Edge Network on Game Theory

Dalian, China

Dalian University of Technology

Dec. 2019 - Jun. 2020

- Realized a decentralized coalition algorithm based on *Gale-Shapley algorithm* to minimize the total latency of VNF scheduling.
- Proved Nash Equilibrium in the algorithm framework and bolstered the robustness of VNF service chains allocation.
- Compared with the existing *Genetic Algorithm* and *Round-robin Scheduling Algorithm*, and the results showed better scheduling time.

Teaching Assistant

Communication Networks

Zurich, Switzerland

ETH Zurich

Feb. 2025 - May 2025

- Developed the frontend development of the course chatbot.
- Conducted three exercise sessions focusing on the reliable transport and the application layer.

Selected Awards

2022	KTH Erasmus+ Scholarship
2021	EPFL Swiss-European Mobility Program (SEMP) Scholarship
2021	KTH Covid-19 Financial Aid Scholarship Based on Academic Excellence
2020	Distinguished Undergraduate of Dalian City
2020	Outstanding Undergraduate Thesis of Dalian University of Technology
2018	Outstanding Youth League Member of Dalian University of Technology

Skills

Programmings	advanced: Python; proficient: JavaScript/HTML/CSS, Java, Scala, C/C++; have experience: TLA+, Go.
Tools	PyTorch, Hadoop, Spark, jQuery, Ajax, Elasticsearch, DPDK, Kubernetes, Helm.
Miscellaneous	Linux, Shell (Bash/Zsh), Clang, CMake, Bazel, LLVM, Git, CI/CD, Jenkins, LaTeX.
Languages	English (TOEFL 101, 2023), Chinese (Native), Japanese (N2), German (A1), French (A1).
Extracurricular	Living and studying in Europe, I've embraced diverse skills. I annually ski in the Swiss Alps and dive in the Mediterranean with an open water diving certificate. Being an amateur photographer is my hobby and my gallery can be found here . Moreover, motivated by my European roommates, I've started exploring various cuisines and honing my culinary skills (photos available here).